AQ5.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Let $T(x, y, z) = mx + ny + pz$ denote the total amount of money a multiplex made from a specific movie, in one day by selling x number of tickets for the morning show, y number of tickets for the evening show, and z number of tickets for the night show, where m, n and p are the prices of each ticket of that movie in the morning, evening, and night show, respectively.

In the multiplex, 2 Malayalam movies, 1 Hindi movie, 1 Bengali movie, and 1 English movie are running parallelly in one week. Suppose there are 3 shows (Morning, Evening, Night) per day for each of the Malayalam movies and the English movie, whereas 2 shows (Evening, Night) per day for the Hindi and the Bengali movie. Ticket price (irrespective of morning, evening, and night shows) for each Malayalam movie is ₹200, Hindi movie is ₹250, Bengali movie is ₹250, and English movie is ₹300. The number of tickets sold in a particular day is given in Table M2W5AQ2:

Answer questions 1 to 8 using the given data above.

Level 1

1 point
Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the morning shows only?

- $200m_1 + 200m_2 + 200m_3$
- $200m_1 + 250m_1' + 300e_1$
- $200m_1 + 200m_1' + 250e_1$
- $200m_1 + 200m_1' + 300e_1$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$200m_1 + 200m_1' + 300e_1$

1 point
Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali movie only?

- $200(b_2 + b_3)$
- $250(b_2 + b_3)$
- $250(h_2 + h_3)$
- $250b_2 + 200b_3$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$250(b_2 + b_3)$250(b2 + b3)

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the night shows only?

○ $200(m_2 + m_2') + 250(h_2 + b_2) + 300e_2$ 200(m2 + m2′) + 250(h2 + b2) + 300e2

○ $200(m_3 + m_3' + h_3 + b_3) + 300e_3$ 200(m3 + m3′ + h3 + b3) + 300e3

○ $200(m_3 + m_3') + 250(h_3 + b_3) + 300e_3$ 200(m3 + m3′) + 250(h3 + b3) + 300e3

○ $200(m_3 m_3') + 250(h_3 b_3) + 300e_3$ 200(m3m3′) + 250(h3b3) + 300e3

No, the answer is incorrect.
Score: 0
Accepted Answers:

$200(m_3 + m_3') + 250(h_3 + b_3) + 300e_3$ 200(m3 + m3′) + 250(h3 + b3) + 300e3

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali and English movie only?

○ $250(b_2 + b_3)$250(b2 + b3)

○ $300e_1 + 250(b_2 + e_2) + 300(b_3 + e_3)$300e1 + 250(b2 + e2) + 300(b3 + e3)

○ $250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$250(b2 + b3) + 300(e1 + e2 + e3)

○ $b_2 + b_3 + e_1 + e_2 + e_3$ b2 + b3 + e1 + e2 + e3

No, the answer is incorrect.
Score: 0
Accepted Answers:

$250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$250(b2 + b3) + 300(e1 + e2 + e3)

Level 2

1 point

What was the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Malayalam movies together?

○ $T(m_1 + m_1', m_2 + m_2', m_3 + m_3')$T(m1 + m1′, m2 + m2′, m3 + m3′)

○ $200T(m_1 + m_1', m_2 + m_2', m_3 + m_3')$200T(m1 + m1′, m2 + m2′, m3 + m3′)

○ $T(m_1 m_1', m_2 m_2', m_3 m_3')$T(m1m1′, m2m2′, m3m3′)

○ $200T(m_1 m_1', m_2 m_2', m_3 m_3')$200T(m1m1′, m2m2′, m3m3′)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(m_1 + m_1', m_2 + m_2', m_3 + m_3')$T(m1 + m1′, m2 + m2′, m3 + m3′)

1 point

Which of the following options are correct?

☐ $T(x,y,z) = T(x,0,0) + T(0,y,0) + T(0,0,z)$

☐ $T(x,y,z) = T(x,y,0) + T(0,0,z)$

☐ $T(x,y,z) = T(x,0,0) + T(y,0,0) + T(z,0,0)$

☐ $T(x,0,0) + T(y,0,0) + T(z,0,0) = T(x+y+z,0,0)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(x,y,z) = T(x,0,0) + T(0,y,0) + T(0,0,z)$

$T(x,y,z) = T(x,y,0) + T(0,0,z)$

$T(x,0,0) + T(y,0,0) + T(z,0,0) = T(x+y+z,0,0)$

1 point

Which of the following equalities correctly represent the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali and Hindi movie together?

☐ $T(0, b_2+h_2, b_3+h_3) = T(0, (h_2+b_2), (h_3+b_3))$

☐ $T(0, b_2+h_2, b_3+h_3) = T(0,0,0) + T(0, h_2+b_2, 0) + T(0, h_3+b_3, 0)$

☐ $T(0, b_2+h_2, b_3+h_3) = T(0, (h_2+b_2), 0) + T(0, 0, (h_3+b_3))$

☐ $T(0, b_2+h_2, b_3+h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(0, b_2+h_2, b_3+h_3) = T(0, (h_2+b_2), (h_3+b_3))$

$T(0, b_2+h_2, b_3+h_3) = T(0,0,0) + T(0, h_2+b_2, 0) + T(0, h_3+b_3, 0)$

$T(0, b_2+h_2, b_3+h_3) = T(0, (h_2+b_2), 0) + T(0, 0, (h_3+b_3))$

$T(0, b_2+h_2, b_3+h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$

1 point

Which of the following equality correctly represents the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the English movie?

○ $T(e_1, e_2, e_3) = 200(e_1+e_2+e_3)$

○ $T(e_1, e_2, e_3) = 250(e_1+e_2+e_3)$

- ○ $T(e_1, e_2, e_3) = 200e_1 + 250e_2 + 300e_3$ T(e1, e2, e3) = 200e1 + 250e2 + 300e3
- ○ $T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$ T(e1, e2, e3) = 300(e1 + e2 + e3)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$ T(e1, e2, e3) = 300(e1 + e2 + e3)

Check Answers

Your score is: 0/8